MIAMI UNIVERSITY

SIGHT Project - Meeting 26

Project Name: Safety Immersion and Gamified Hazard Training for Industry 5.0 Workers

Project Number: WSIC26-250206-009

Date: October 02, 2026

Time: 11:00 AM EDT

Location: Zoom

Meeting Focus: Recap of Meeting 25 action items, administrative updates (Q5 report, project meeting schedule, outreach), SIGHT-VR v1.0 and the Makino training demonstrations, the first SOP-guided manual mill run, MaxByte's annotation progress, and a closed session on AR environment status and the experimental protocol.

Attendees

- **Miami University (MU):** Fadel Megahed (PI), Mohammad Mayyas (Co-PI), Jay Shan (Co-PI), Ibrahim Yousif (Associate Research Scientist), Michael Wise (GRA), Austin Hamilton (GRA/SIGHT AR Lead), Lindsay Stevenson (Student Research Assistant), Leo Lyublinski (Student Assistant), Mo Farrag (Project Manager)
- **Discovery Center:** Kristen Morio (Evaluator), Yue Li (Evaluator), Abby Helsinger (Senior Research Associate/Evaluator)
- **External Consultant:** Lora Cavuoto (University of Buffalo)
- **MaxByte:** Ramshankar C S (CEO)
- **Industrious Group Inc.:** Jeff Malek (Corporate Director of EHS)

1. Introductory Remarks & Recap of Meeting 25

- **Welcome & Recap:** Fadel Megahed opened the meeting by summarizing the four weeks since Meeting 25 (September 4, 2026). The Meeting 25 minutes were circulated and shared with the BWC; no corrections were raised.
- **Introduction:** Lindsay Stevenson (MSBA) introduced herself. Lindsay will focus on AI evaluation, in particular assessing the faithfulness of the AI assistant's answers relative to the safety manual.
- **Status of Meeting 25 Action Items:**
    - Move annotations to RoboFlow: done for the CNC lathe and cobot; MaxByte can share the RoboFlow export and is converting the output to YOLO format.

- Secure the Magic Leap 2 headset: done. The backup headset has been received and is working as expected (Austin).
- Provide the Meeting 25 decks and summaries to the BWC: done.
- Q5 report draft: in progress; Fadel plans to share a partial draft on Monday (October 5) and a full draft on Tuesday (October 6).
- Computer vision model for Austin (Ibrahim): not yet delivered; Fadel will follow up.

2. Administrative Updates

- **BWC Deliverables Status:**
  - The Q5 report (DO7.4) is due October 9, 2026.
  - **Project meeting schedule (Deliverable 2.1):** The schedule submitted in 2025 listed September 11 as the September meeting, but the team met on September 4. After checking with the BWC, Fadel updated Deliverable 2.1 so that September 4 is the reported September meeting. The November meeting was also corrected to November 13, since no meeting is planned for November 6. Meetings reported to the BWC must match Deliverable 2.1.
- **Farmer School Dissemination Session:** Arthur proposed presenting SIGHT to Farmer School and ISA faculty and leadership. The session is scheduled for October 9, 2026, 12:00 to 1:30 PM, in room 2046, with lunch provided; Arthur will lead it. Team members are encouraged to attend.
- **Industrious Group Visit:** Jeff Malek visited on Friday, September 25, for about three hours (breakfast and a tour of the AM Hub labs led by Michael). Jeff was able to put on the AR headset and start the experience. Lesson learned: the team needs a short onboarding process for first-time AR users; Austin and Michael are working on it.
  - *Next Step:* A team visit to Industrious Group in Cleveland. Jeff is targeting early November and expects to confirm early next week. Fridays work best for the MU team, and a visit before winter is preferred.
  - *Documentation:* No photos were taken during the September 25 visit; the team will document the Cleveland visit.
  - Fadel thanked Jeff for his feedback and for his email suggesting conferences for the team.
- **ASSP Safety 2027 Conference:** Fadel and Lora submitted their presentation proposal. Lora expects a decision around November.

3. Technical Updates: SIGHT-VR, Makino Trainings, and Manual Mill SOP

- **SIGHT-VR Version 1.0 (Arthur, presented by Fadel):**
  - Arthur released version 1.0 of the web-based VR, which the team considers production-ready after tracking and resolving the 85 issues reported at Meeting 25.
  - Changes since Meeting 25 include separating assessment from training completion, adding CNC and PPE assessment quizzes, dashboard improvements, and fixes to the AI assistant.
- **Scaling to New Machines (Makino):**
  - Two new trainings were produced for AM Hub machines: the Makino D200Z 5-axis machining center and the Makino UX3i wire EDM. Each follows the structure of the original modules (six guided modules from setup and safety through cleanup and shutdown). A narrated video of both was shown.
  - The first three machine trainings each took four to six months to reach a working module. With the team's AI-based pipeline, the D200Z training was produced overnight and the UX3i training in about two days. Both are self-initiated demonstrations, and Makino has been contacted for feedback.
  - *Discussion:* The trainings are meant to precede, not replace, hands-on learning in the lab (similar to simulation before driving in driver education), and will be followed by AR-based instruction. The team is discussing how to reduce the steps that still need human intervention.
- **First SOP-Guided Manual Mill Run (Michael, Jay):**
  - On Tuesday, September 29, Jay Shan, who had no prior manual machining experience, ran the manual milling machine using Michael's SOP.
  - Michael noted that Jay used hands-on feel to guide his passes, which offered a useful novice perspective. Mohammad noted that a structured, clear SOP gave the user confidence and supports faster, easier, and safer training.
  - Jay identified items in the SOP to tighten before it is used in AR.

4. Industry Partner Collaboration

- **MaxByte Update (Ramshankar):**
  - More than 9,000 CNC lathe and cobot images were extracted, cleaned, classified per the SOPs, uploaded to RoboFlow, and annotated with polygon boundaries that follow each component's shape (e.g., cobot gripper, controller box, pendant, CNC door, spindle, control

buttons). MaxByte reported an accuracy of about 90 to 95% and is preparing a validation report.
    - *Next Steps:* Annotate the milling machine (with support from Mohammad and the team), export the datasets in YOLO format, and bring them into Unity for the Magic Leap. MaxByte's target is October 23, 2026 for all three machines in AR with annotations.
- **Discussion:**
    - Ibrahim asked why some components in the images are not annotated (e.g., door handles, several control-panel buttons, the robot arm). Ramshankar said the critical components were selected based on the SOPs and that more can be added. MaxByte will share a table of SOP objects versus annotated objects with coverage percentages.
    - Fadel asked that the accuracy figures state how they are calculated (e.g., per image, the number of target objects, how many were captured correctly, and any extra detections).
    - Ibrahim noted that the annotation of the top-down camera images of the cobot environment (cobot, human, and CNC) has been pending for over a month and asked Ramshankar to review the email thread directly. Ramshankar will clarify the intended use of that camera view and respond.

(Note: Industry partners Jeff Malek and Ramshankar C S left the meeting prior to the closed session. The Discovery Center evaluation team stayed for the closed session.)

5. Closed Session: AR Status and Experimental Protocol

- *Next Steps:* Lora will work with Jay on an IRB amendment for screen capture and recording. The team will finalize the protocol soon with Lora and the Discovery Center, and run pre-pilots with team members as participants to train the student experimenters and refine the protocol.
- Kristen Morio noted that she serves on the IRB committee and offered to help with IRB and human subjects questions.

6. Risk Mitigation and Log Updates

- **Hardware Obsolescence (Magic Leap 2):** The backup headset logged at Meeting 25 has been received and tested.
- **IRB Amendment:** Recording, screen capture, and any monitor change require an IRB amendment before testing.

7. Further Discussion and Wrap-Up

- Ibrahim asked that the AM Hub team be added to the agenda for the next biweekly update.
- The next meeting is in two weeks.

Action Items

Task	Responsible	Deadline
Share the Q5 report draft (partial, then full)	Fadel Megahed	October 5 / October 6, 2026
Lead the Farmer School dissemination session	Arthur Carvalho	October 9, 2026
Confirm the date for the Cleveland visit	Jeff Malek	Week of October 5, 2026
Develop onboarding for first-time AR headset users	Austin Hamilton / Michael Wise	Before the Cleveland visit
Annotate the milling machine and deliver all three machines in AR (YOLO export, Unity)	Ramshankar / MaxByte (with Mohammad Mayyas)	October 23, 2026
Renew the Unity license	Ramshankar / MaxByte	ASAP
Share a table of SOP objects versus annotated objects, and explain how accuracy is calculated	Ramshankar / MaxByte	Before the next presentation
Review the email thread on top-down camera annotations and respond	Ramshankar / MaxByte	ASAP
Provide the computer vision model to Austin	Ibrahim Yousif (follow-up: Fadel Megahed)	Carried over

Verify the safety-zone circle stays centered after re-anchoring	Austin Hamilton	Next update
Store AR evaluation data with the VR backend and create a unique participant ID across VR and AR	Austin Hamilton / Arthur Carvalho	Before pre-pilots
Prepare an IRB amendment for screen capture and recording	Lora Cavuoto / Jay Shan	ASAP
Finalize the experimental protocol and schedule pre-pilots	Fadel Megahed, Lora Cavuoto, Discovery Center	ASAP
Add the AM Hub team to the next biweekly update agenda	Fadel Megahed	Next meeting